Axel Orellana

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PROFESSIONAL SUMMARY

Computer Engineering student (B.S., May 2027) with concentrations in AI & Robotics, hands-on digital design experience in Verilog and VHDL on FPGA hardware, Python drone programming, and nearly three years of full-stack development. Bilingual (English/Spanish) leader with a record of coordinating projects, events, and volunteers. Pursuing robotics and digital design roles starting summer 2027.

EDUCATION

B.S. Computer Engineering, *University of the District of Columbia* — Expected May 2027

- **Concentrations:** AI & Robotics; Logistics & International Trade
- **Relevant coursework:** Digital Design, Digital Integrated Circuits, Analog Electronics, Computer Architecture, Cyber-Physical Systems

PROJECTS

FPGA Digital Design *(Coursework)* | *Verilog, VHDL, Vivado, Nexys4 DDR*

- Designed gate-level and data-flow full adders, latches, flip-flops, and LED control logic in both Verilog and VHDL.
- Programmed designs onto physical FPGA boards in Vivado and confirmed matching behavior in both languages.

Line-Following Drone *(Team Project)* | *Python, DJI Tello SDK*

- Programmed a DJI Tello drone in Python to follow a line, with video streaming, basic object detection and avoidance, and a modular code architecture.

Senior Capstone *(In Progress, Team Project)* — Spring 2027

- Team project combining embedded systems, hardware and software, and AI and robotics. Details to be announced.

Home Server and Small-Business Web Build *(Ongoing)* | *Docker, Ubuntu, HTML, CSS, JavaScript*

- Building a self-hosted Ubuntu server with Docker and a website plus software infrastructure for a small welding business.

EXPERIENCE

Junior Coder / Full-Stack Developer, *Limbic Systems (CareerWise DC Apprenticeship)* — Oct 2023 – Aug 2026

- Built front-end and back-end systems for small businesses and local governments, including databases and AI features.
- Developed MVPs and AI agent prototypes with Bubble and Botpress through API integration.

Junior PM Coordinator, *UDC Office of Information Technology* — May 2025 – Dec 2025

- Supported project coordination for the university's Office of Information Technology.

PM Host, *Boqueria Dupont (Part-Time)* — Current

LEADERSHIP

President, *Corazón Latino Coalition (formerly Next-Gen Leaders Association), UDC* — May 2025 – Present

- Organized the "Advocacy 101" immigration event with PSSA and the "Ida B. Free" criminal justice workshop with Harriet's Wildest Dreams.
- Served as a panelist at an Urban Institute event on apprenticeship equity and represented UDC at the Spanish-language Camino Al Futuro event.

Candidate, USGA Presidential Campaign, *UDC* — 2026

- Coordinated campaign volunteers, wrote campaign messaging, and managed compliance with school election rules.

TECHNICAL SKILLS

Programming and HDL: Python, Java, JavaScript, HTML, CSS, Verilog, VHDL

Frameworks and Platforms: React, Node.js, Bubble, Botpress, Docker, Ubuntu

Hardware and Tools: Vivado, FPGA (Nexys4 DDR), Raspberry Pi, Arduino, MATLAB, Mbed Studio, Embedded Systems

Languages: English, Spanish